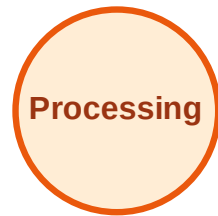


BFS Traversal

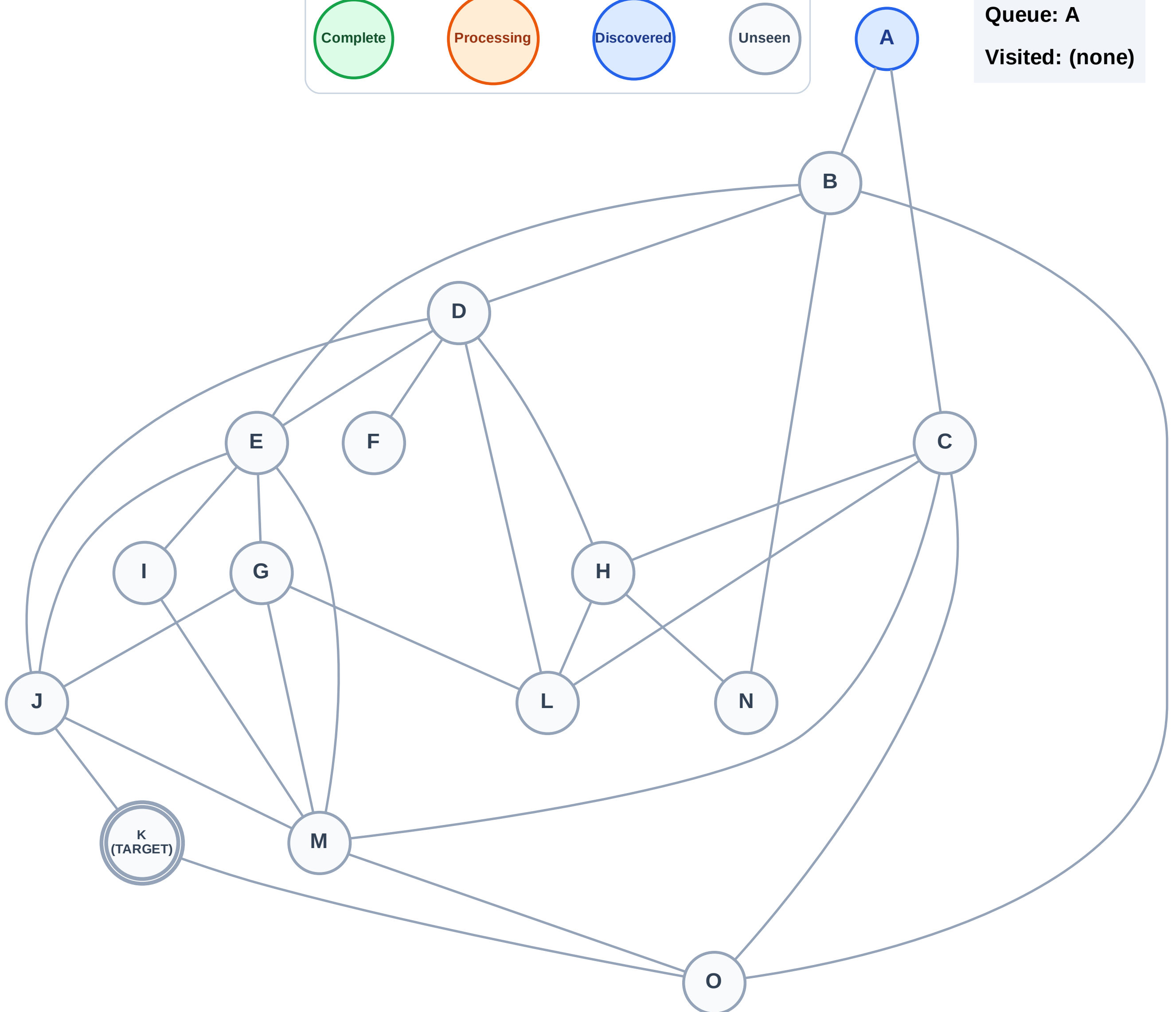
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Queue: A

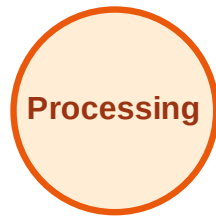
Visited: (none)



BFS Traversal

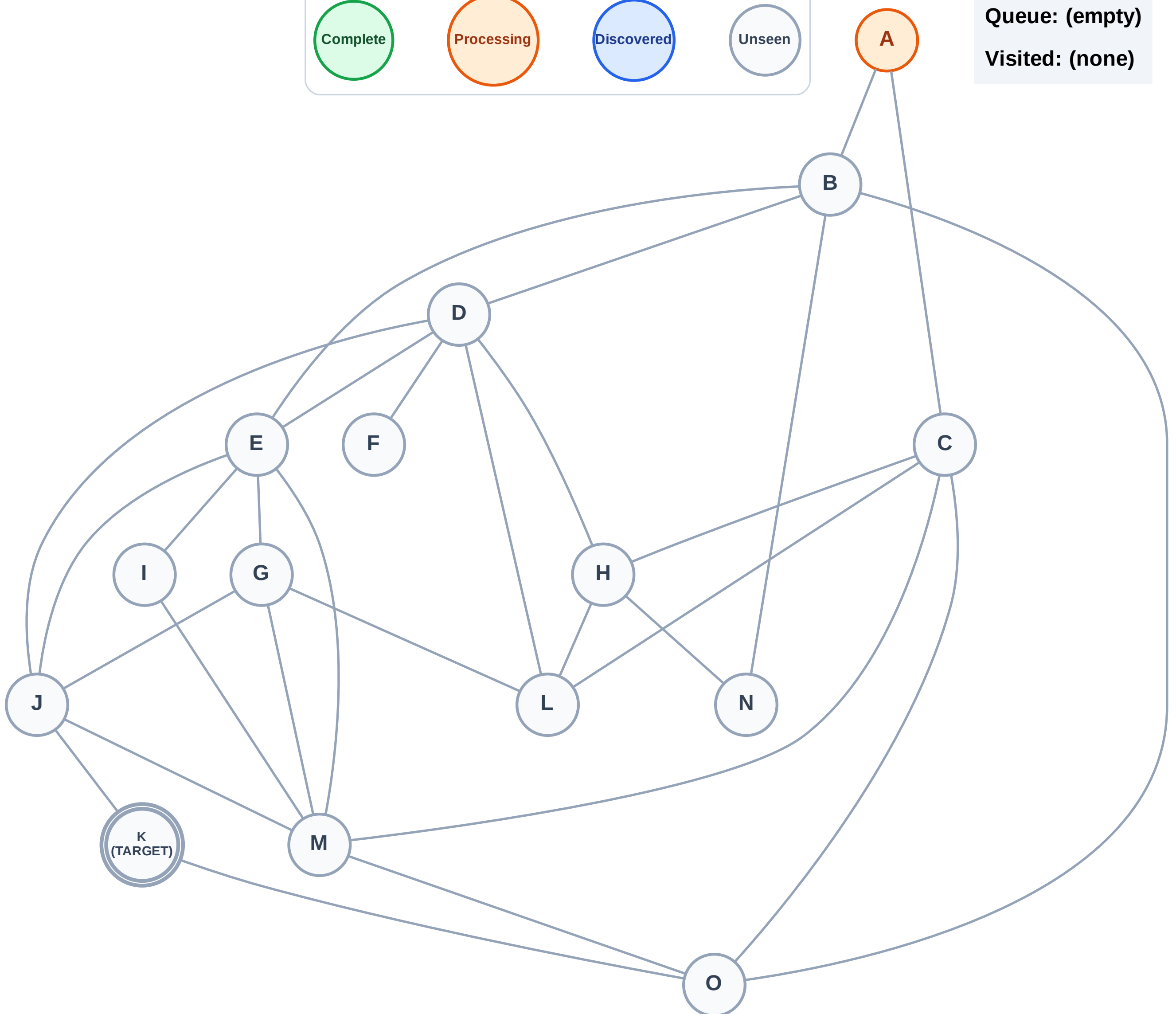
Step 2: Process node A

Legend



Queue: (empty)

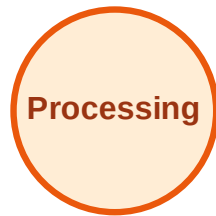
Visited: (none)



BFS Traversal

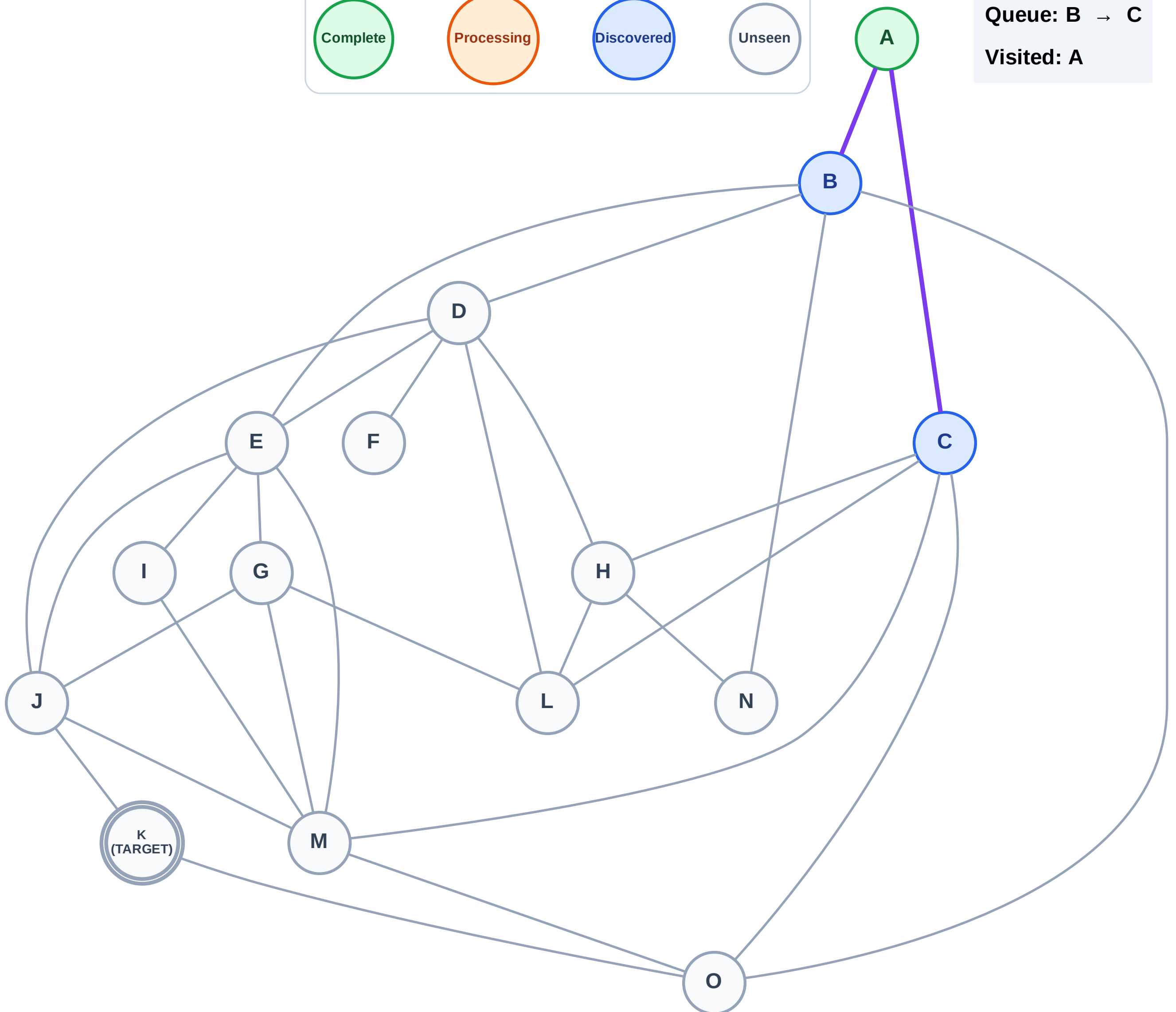
Step 3: Discover B, C from A

Legend



Queue: B → C

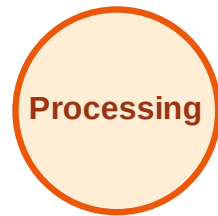
Visited: A



BFS Traversal

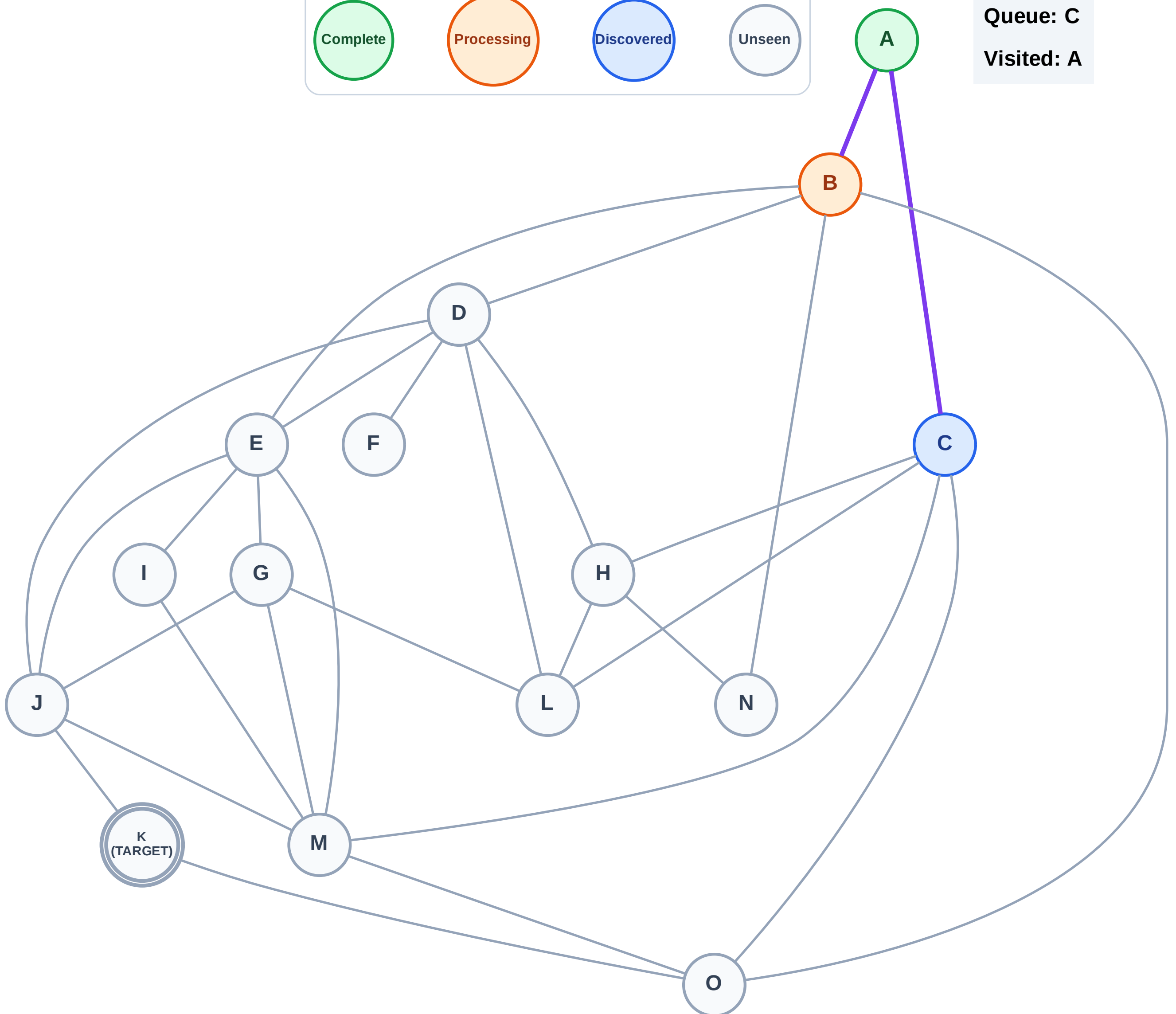
Step 4: Process node B

Legend



Queue: C

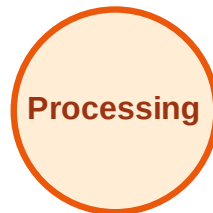
Visited: A



BFS Traversal

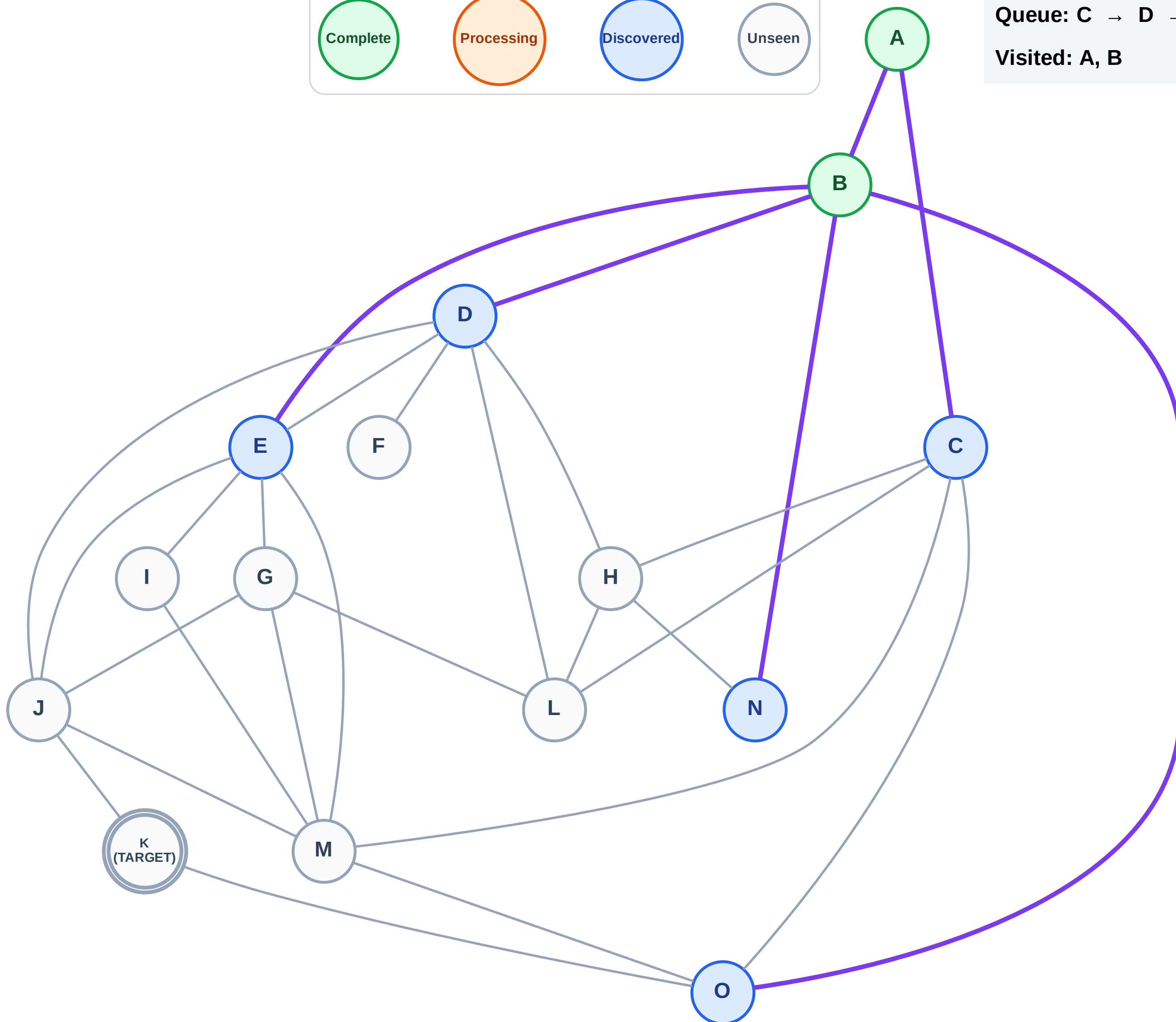
Step 5: Discover D, E, N, O from B

Legend



Queue: C → D → E → N → O

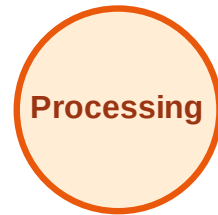
Visited: A, B



BFS Traversal

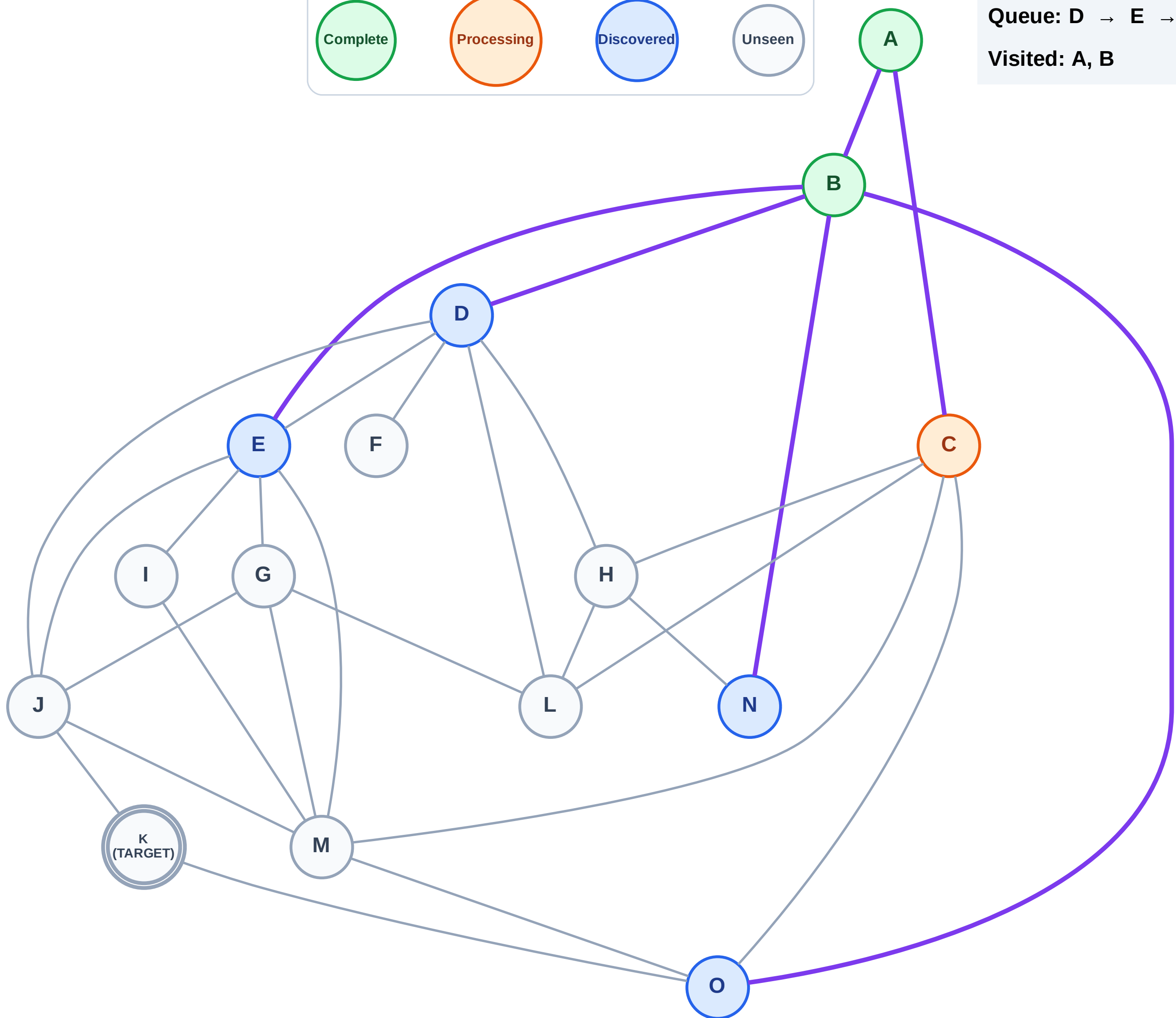
Step 6: Process node C

Legend



Queue: D → E → N → O

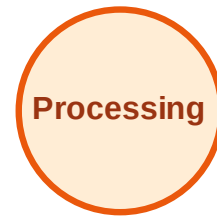
Visited: A, B



BFS Traversal

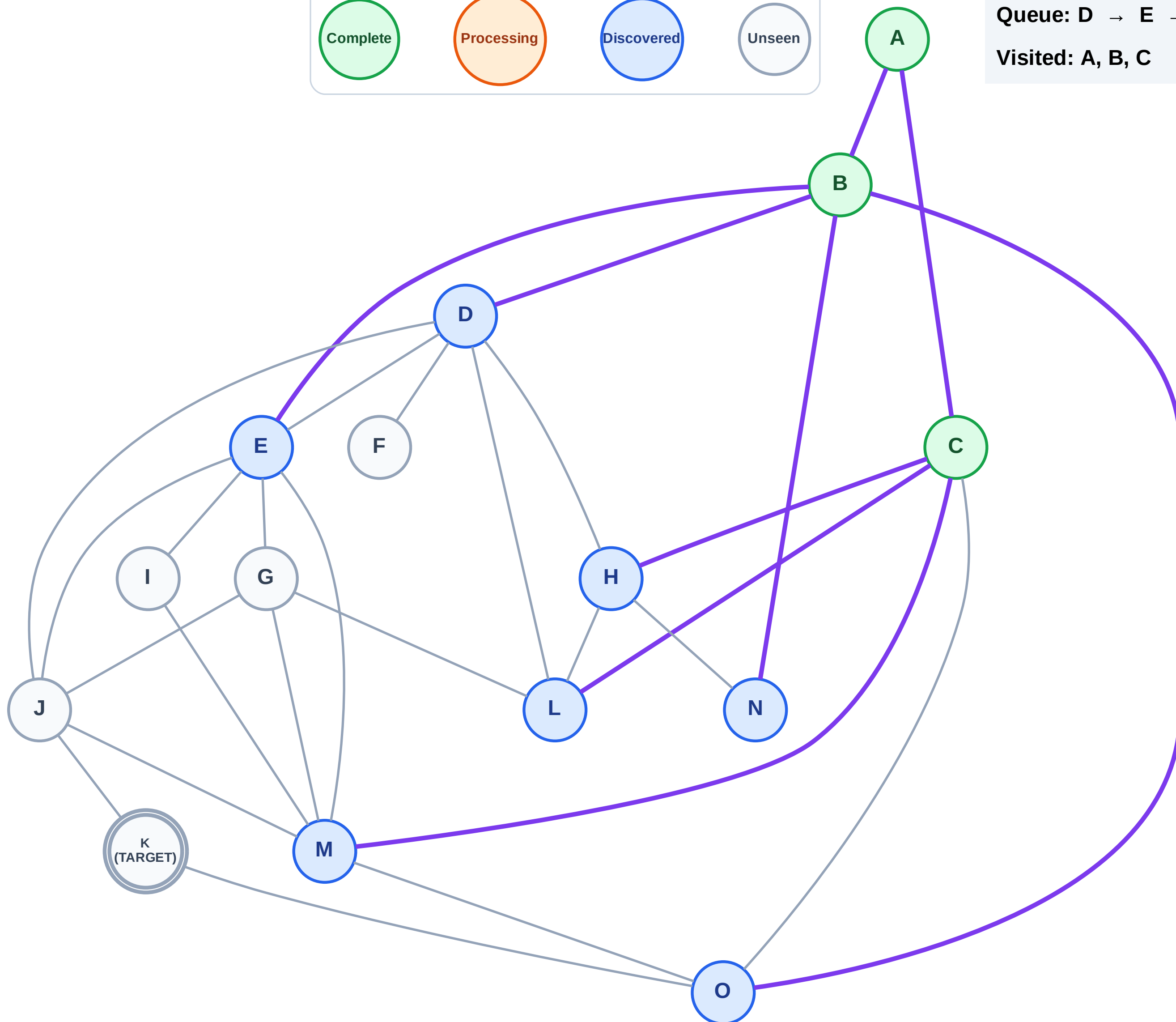
Step 7: Discover H, L, M from C

Legend



Queue: D → E → N → O → H → L → M

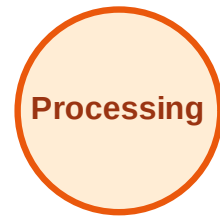
Visited: A, B, C



BFS Traversal

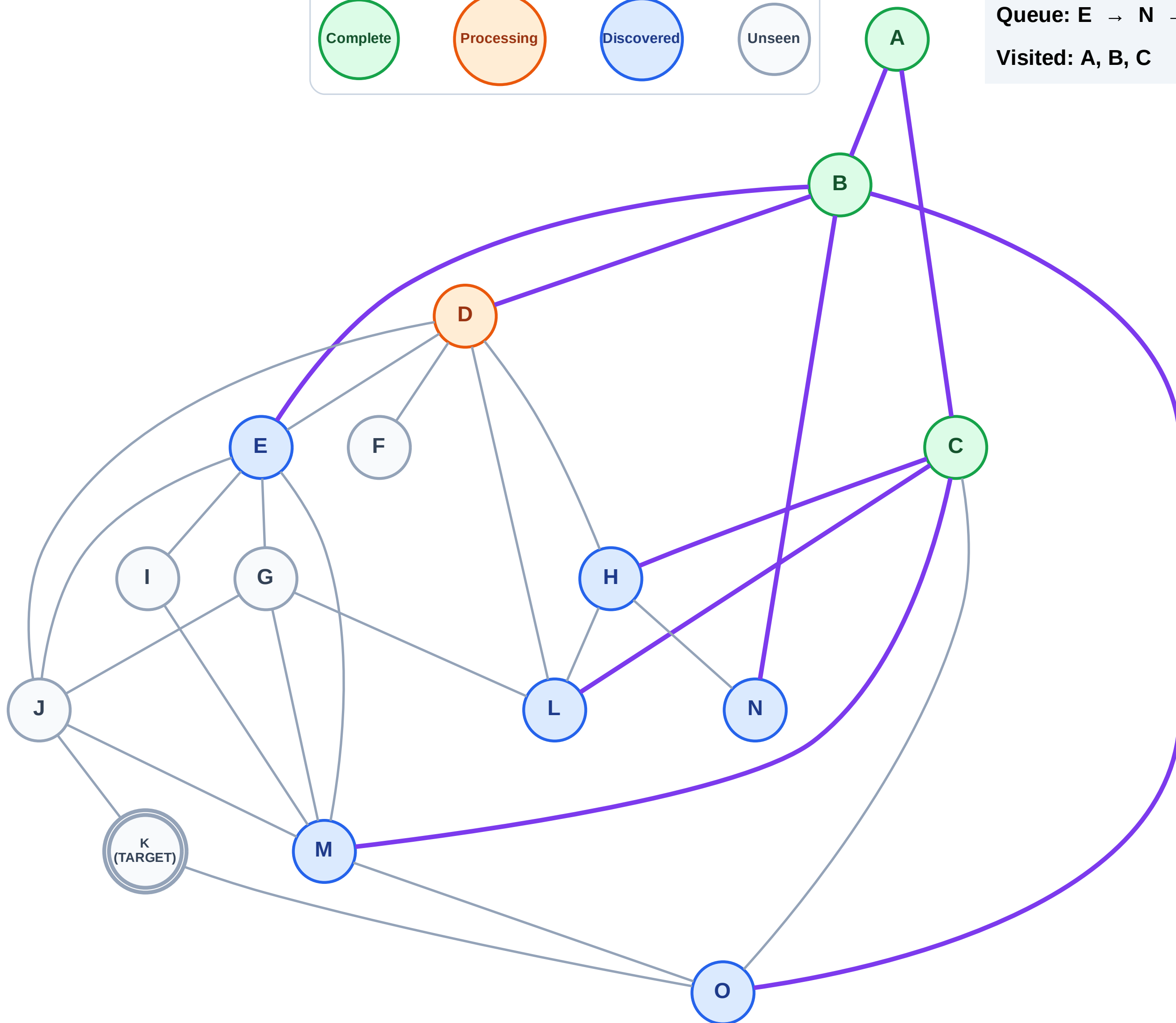
Step 8: Process node D

Legend



Queue: E → N → O → H → L → M

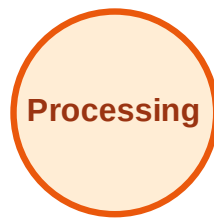
Visited: A, B, C



BFS Traversal

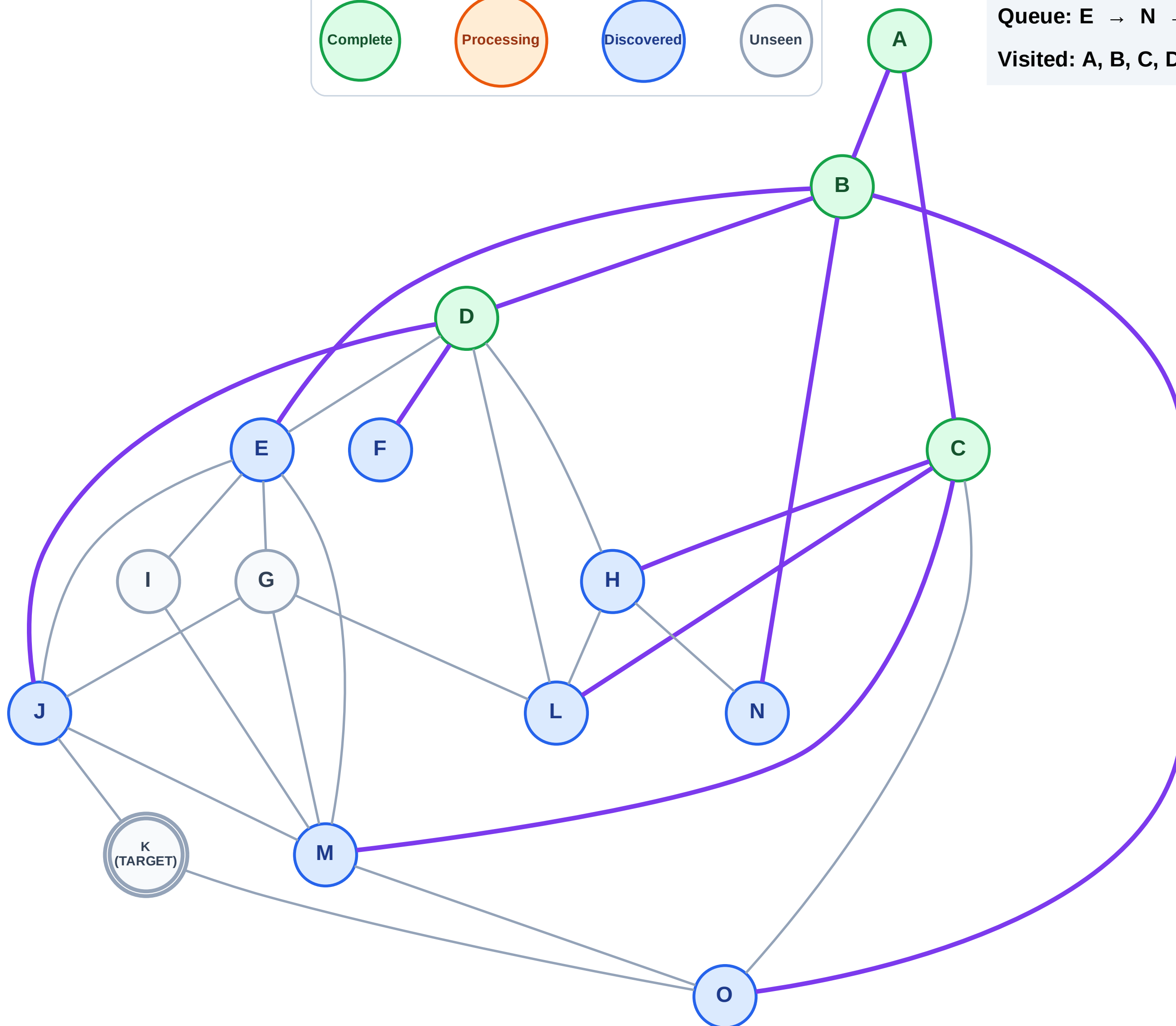
Step 9: Discover F, J from D

Legend



Queue: E → N → O → H → L → M → F → J

Visited: A, B, C, D



BFS Traversal

Step 10: Process node E

Legend

Complete

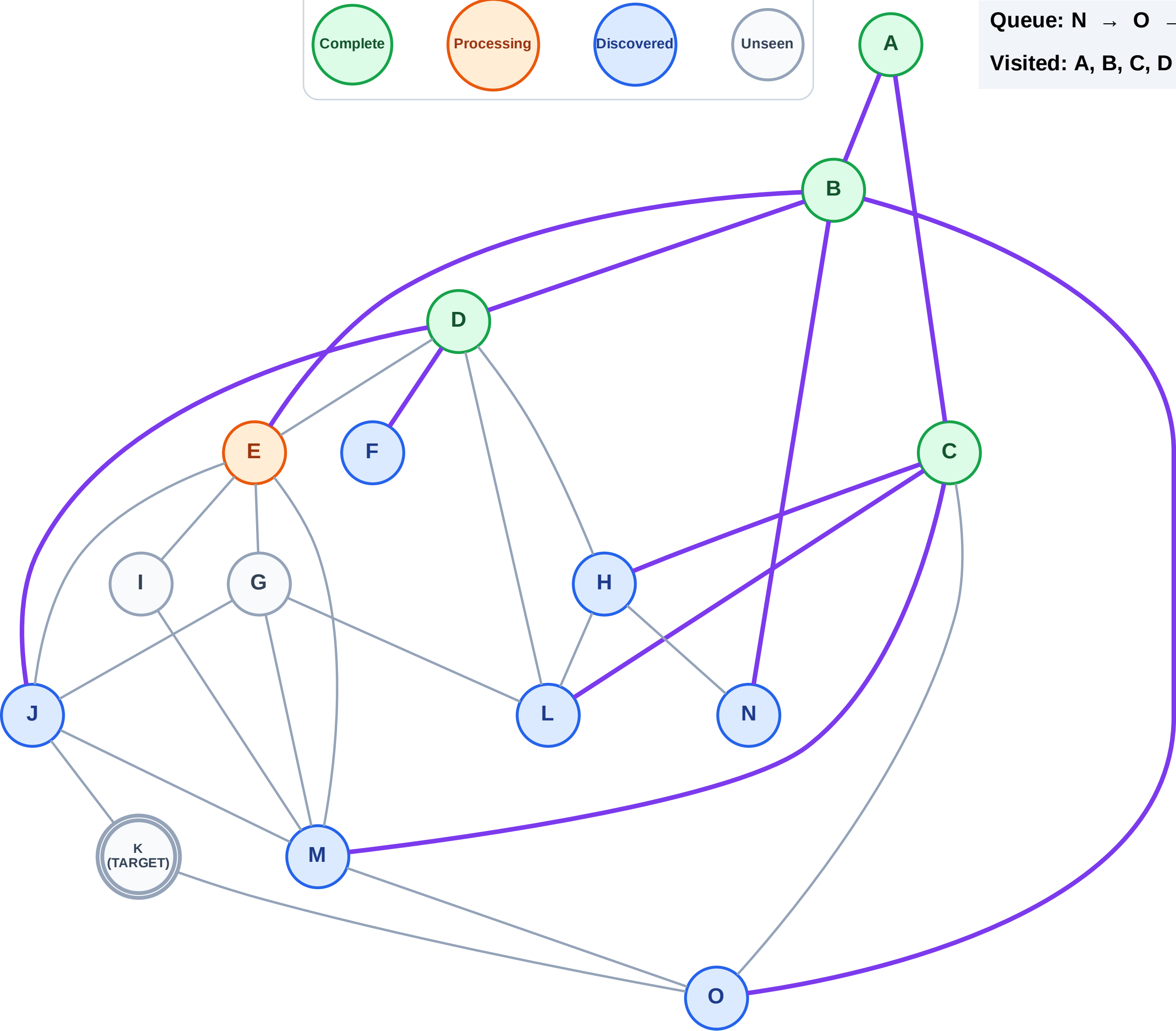
Processing

Discovered

Unseen

Queue: N → O → H → L → M → F → J

Visited: A, B, C, D



BFS Traversal

Step 11: Discover G, I from E

Legend



Complete



Processing



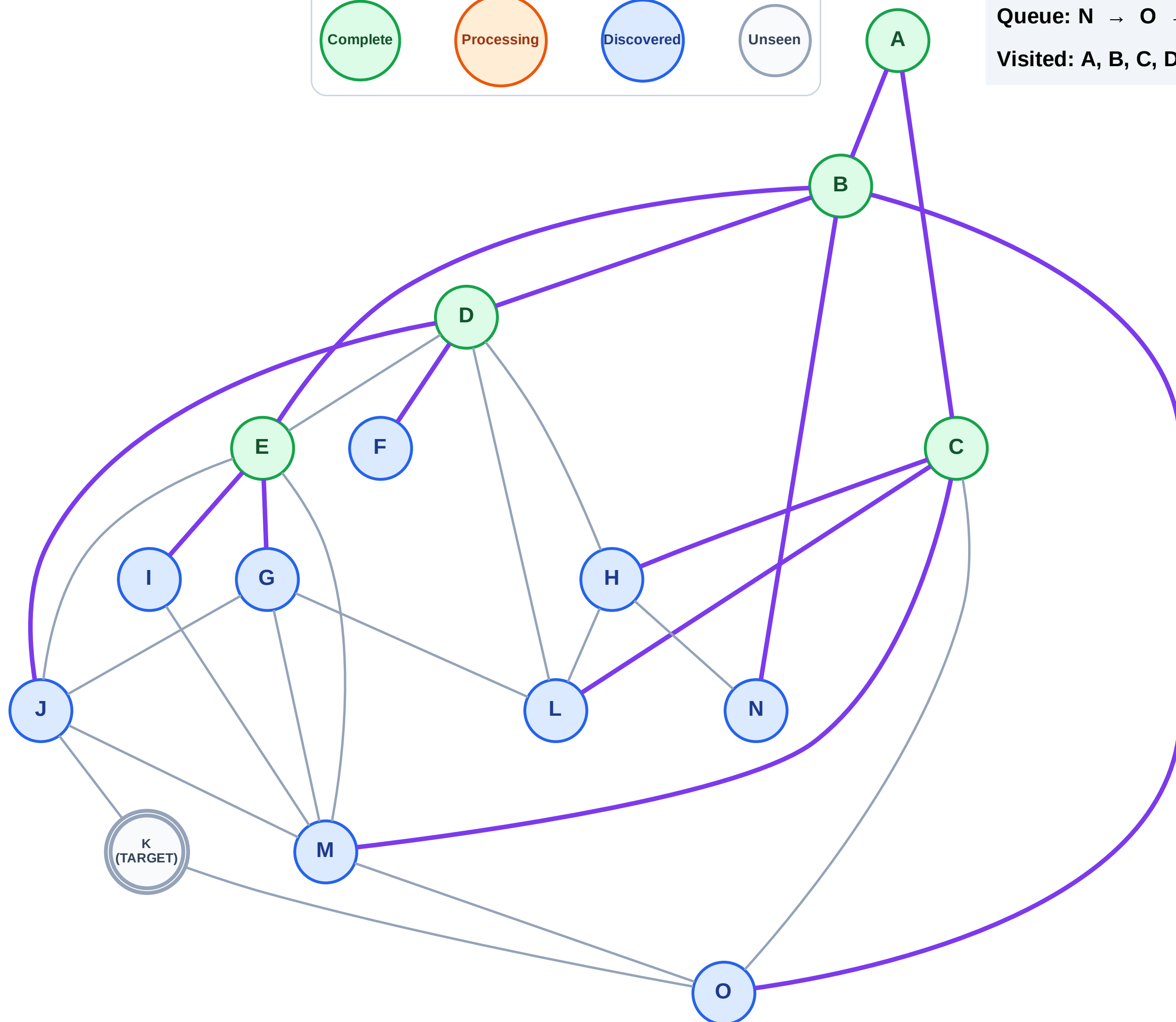
Discovered



Unseen

Queue: N → O → H → L → M → F → J → G → I

Visited: A, B, C, D, E



BFS Traversal

Step 12: Process node N

Legend

Complete

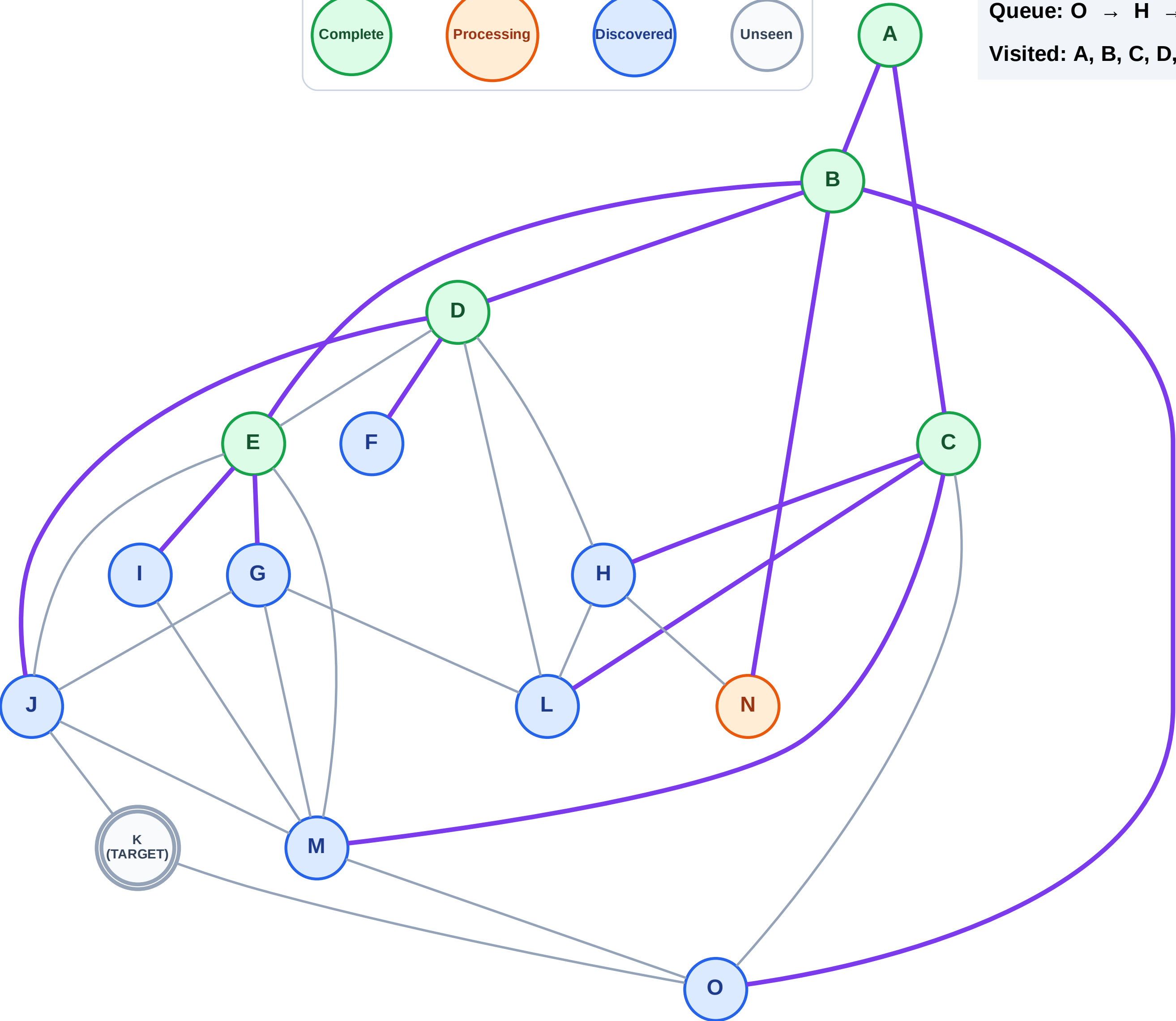
Processing

Discovered

Unseen

Queue: O → H → L → M → F → J → G → I

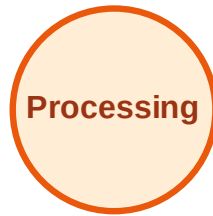
Visited: A, B, C, D, E



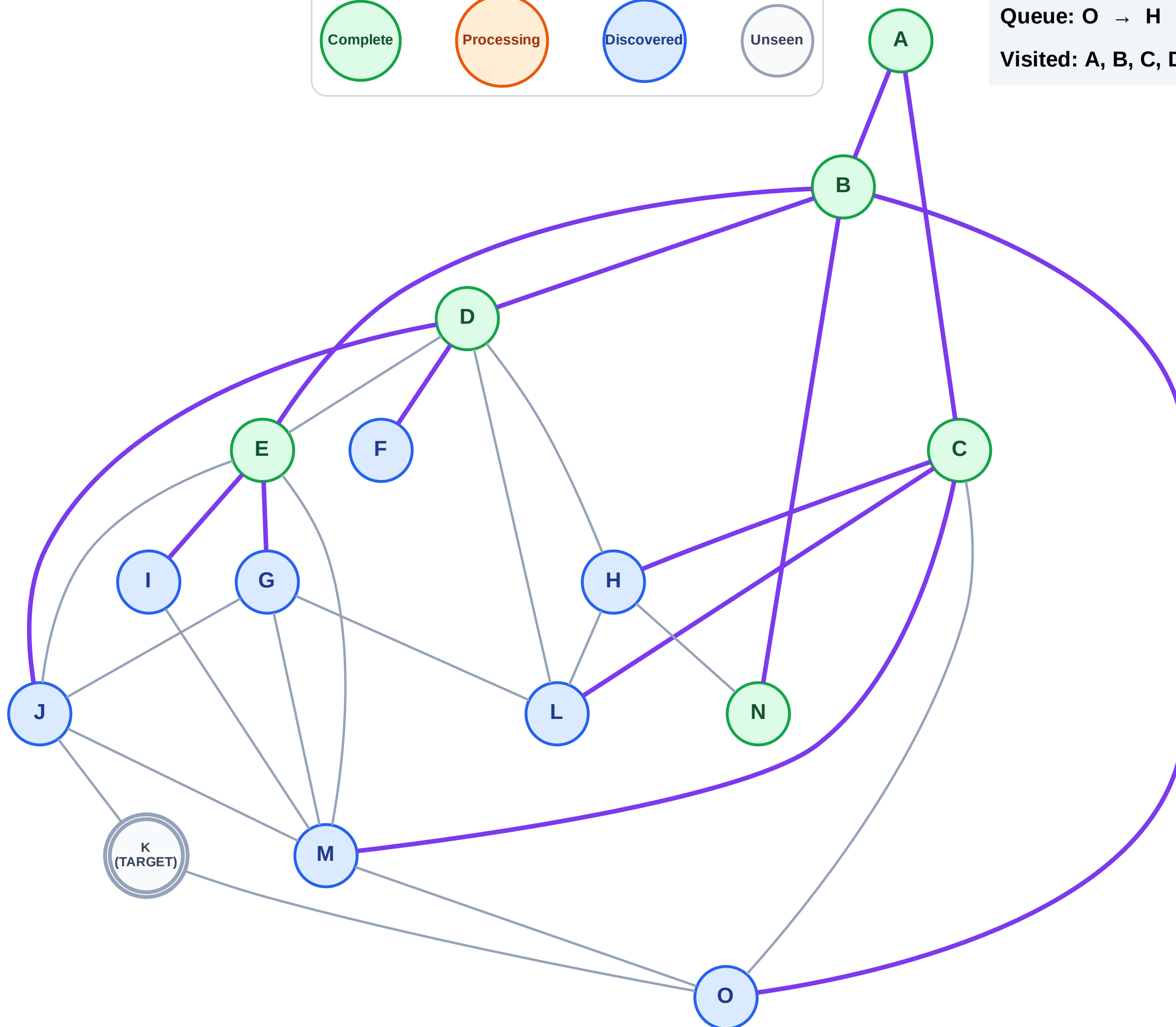
BFS Traversal

Step 13: No new neighbors from N

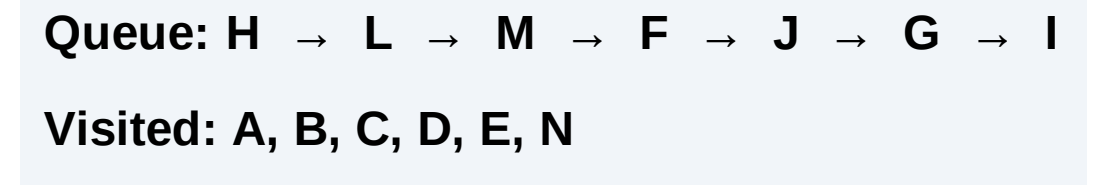
Legend



Queue: O → H → L → M → F → J → G → I
Visited: A, B, C, D, E, N



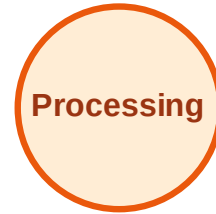
Step 14: Process node O



BFS Traversal

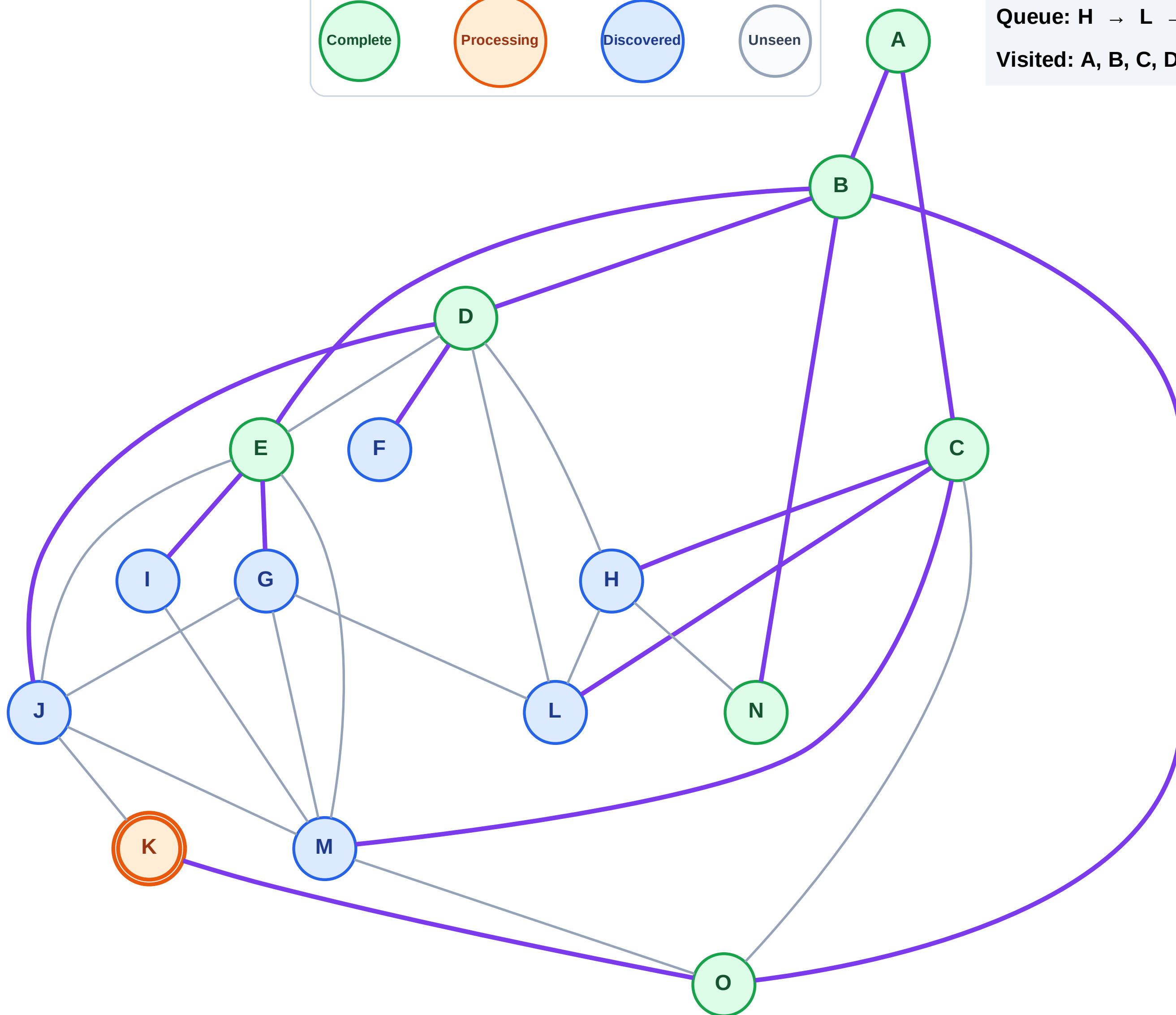
Step 15: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → O → K

Legend



Queue: H → L → M → F → J → G → I

Visited: A, B, C, D, E, K, N, O



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

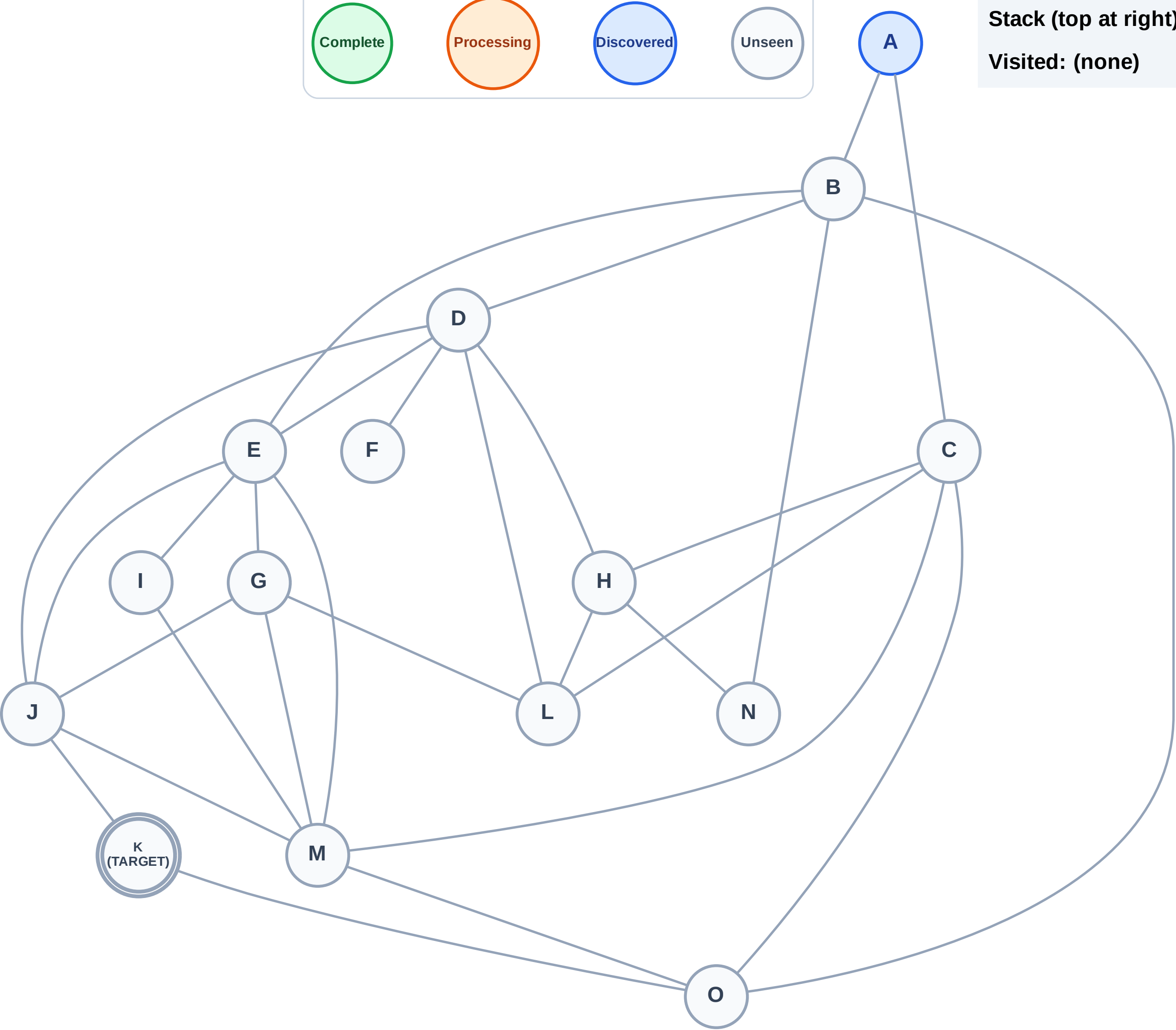
Complete

Processing

Discovered

Unseen

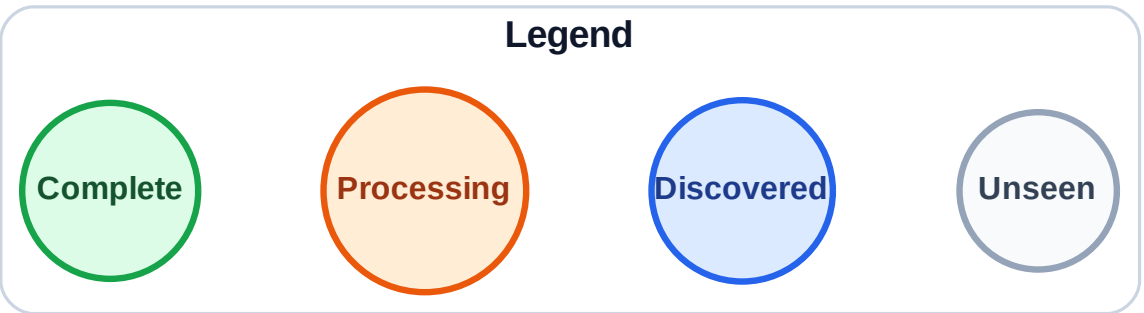
Stack (top at right): A
Visited: (none)



DFS Traversal

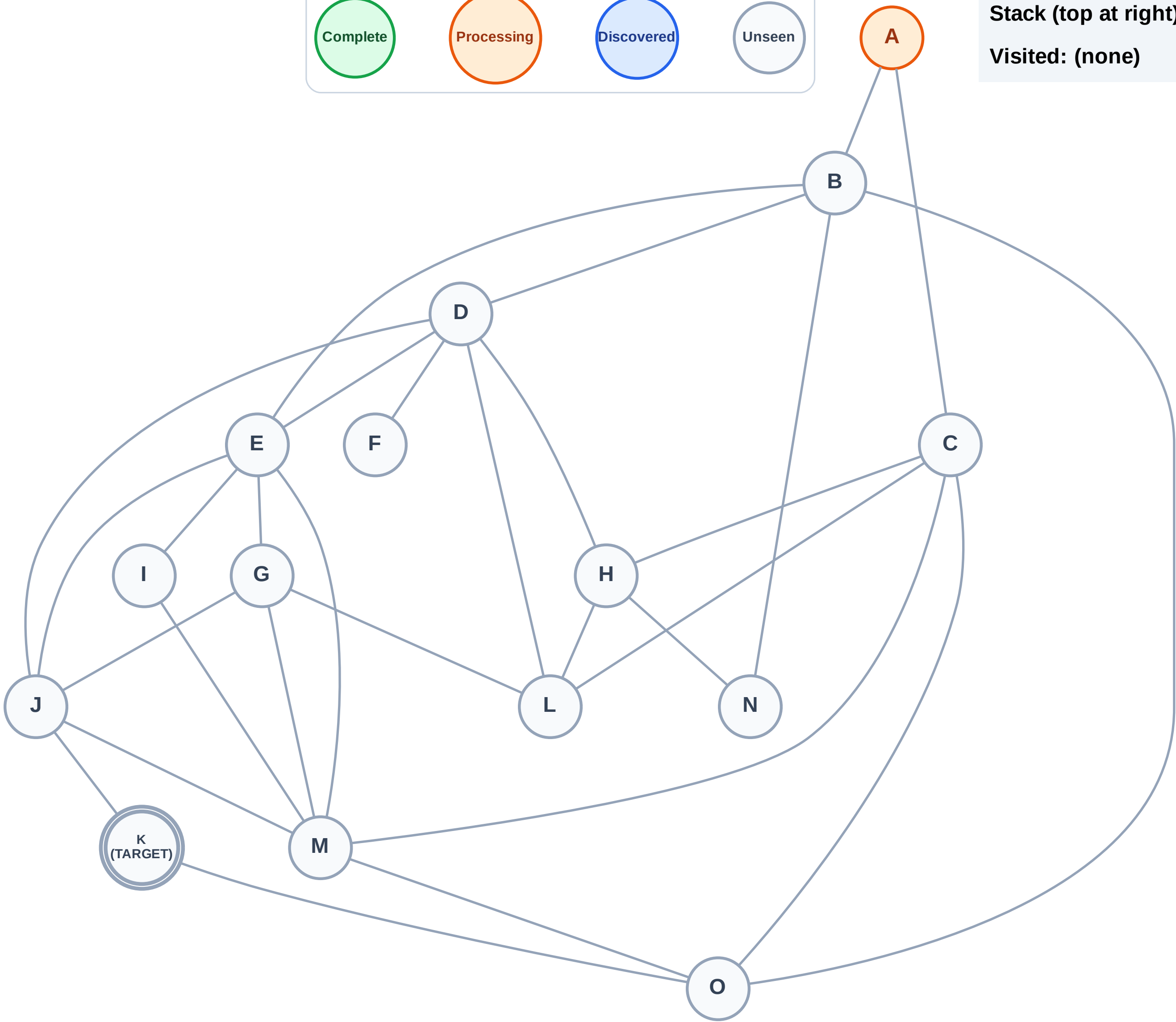
Step 2: Process node A

Legend



Stack (top at right): (empty)

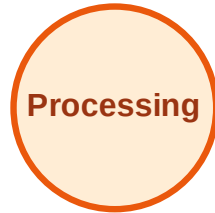
Visited: (none)



DFS Traversal

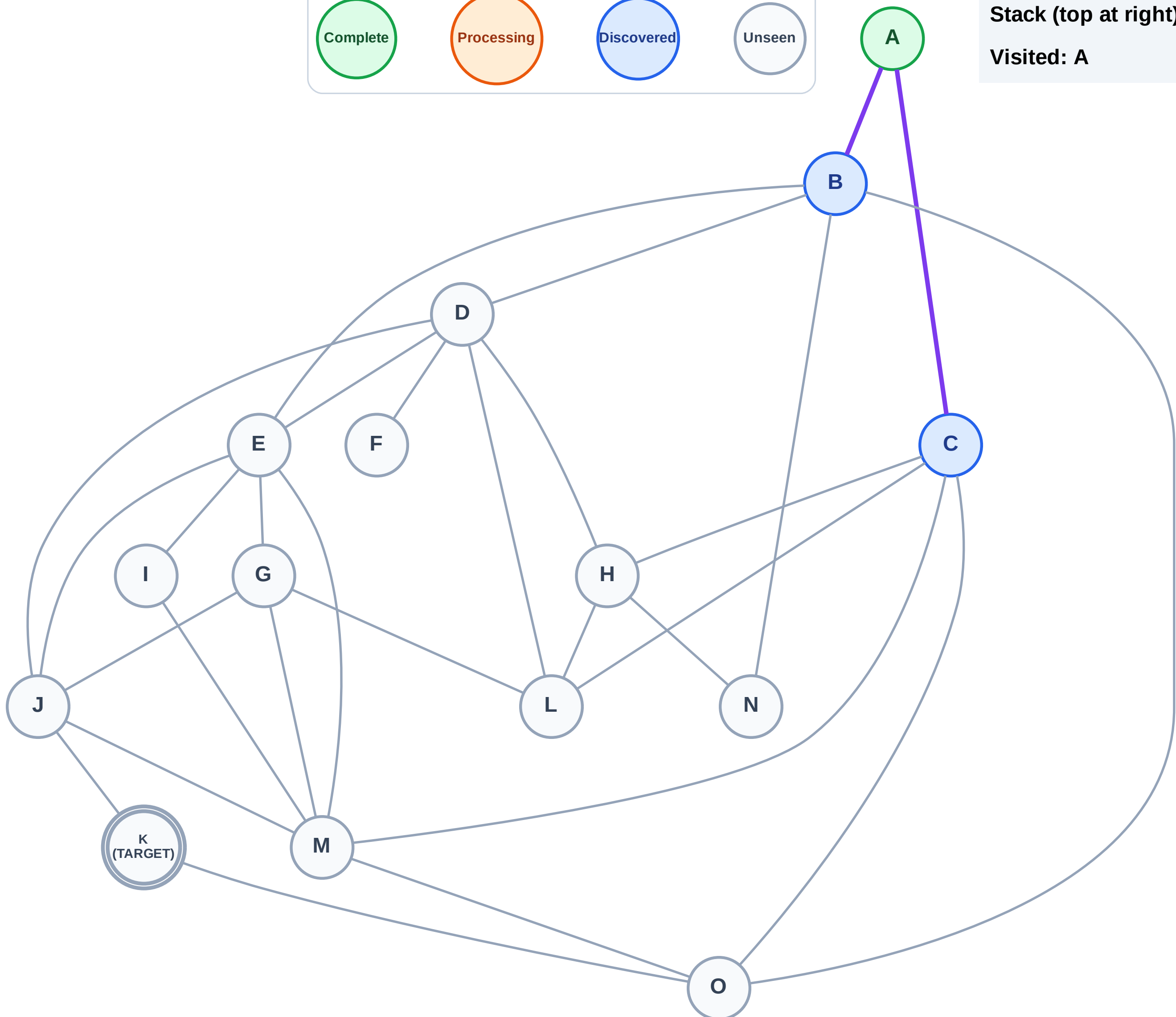
Step 3: Discover C, B from A

Legend



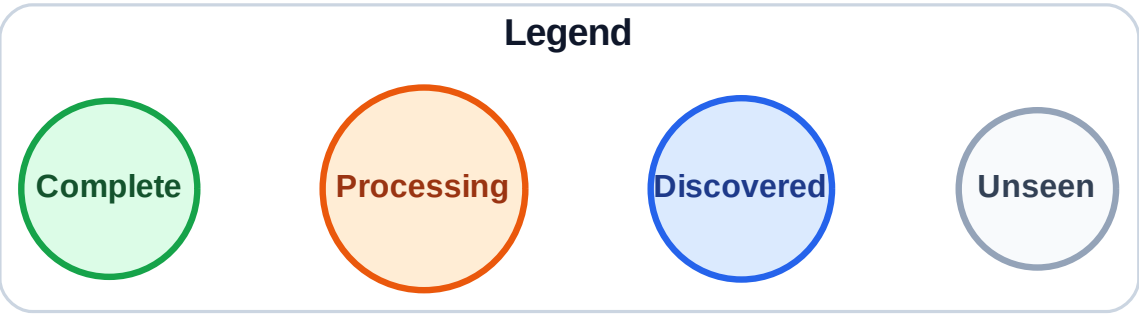
Stack (top at right): C | B

Visited: A



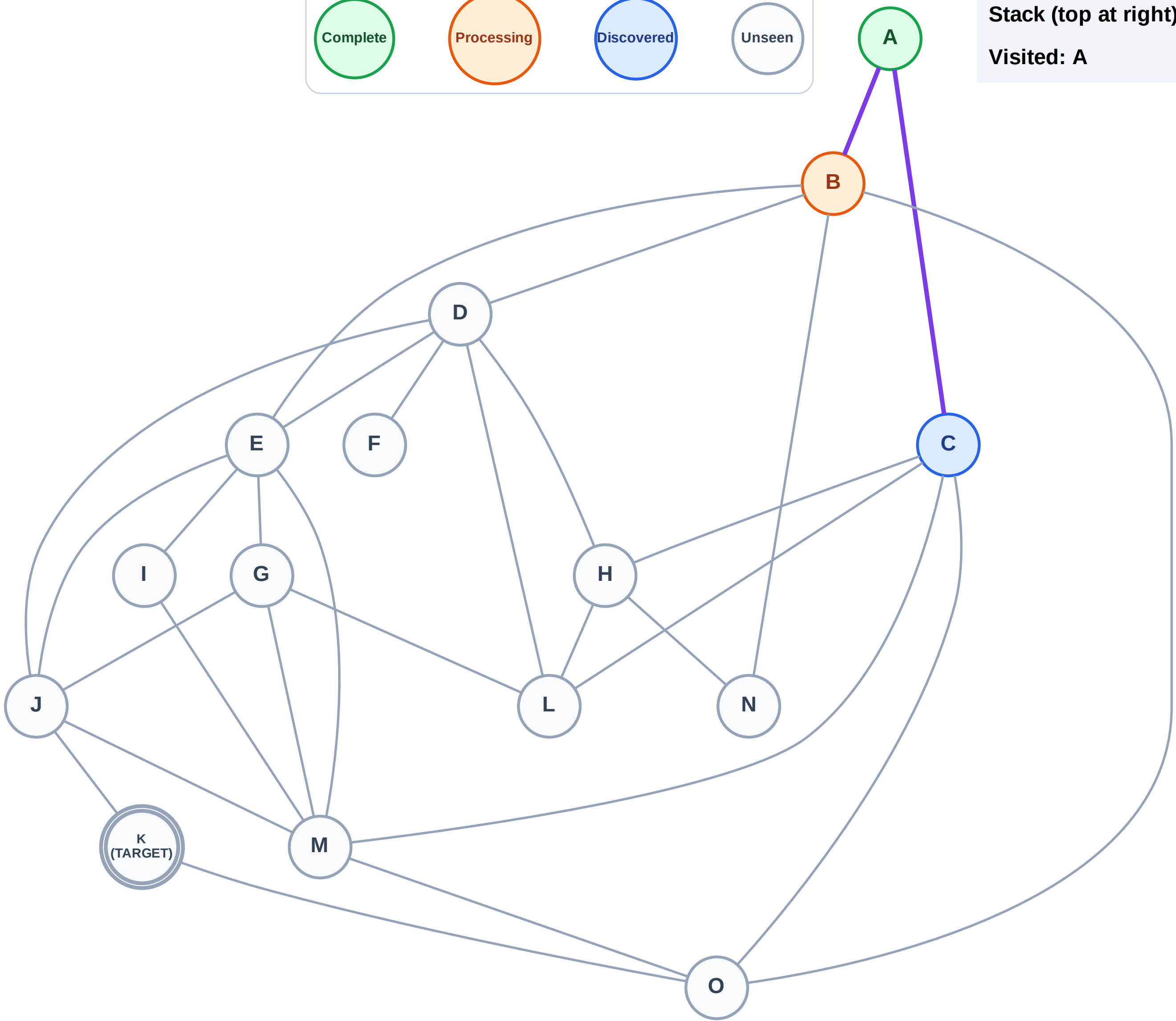
DFS Traversal

Step 4: Process node B

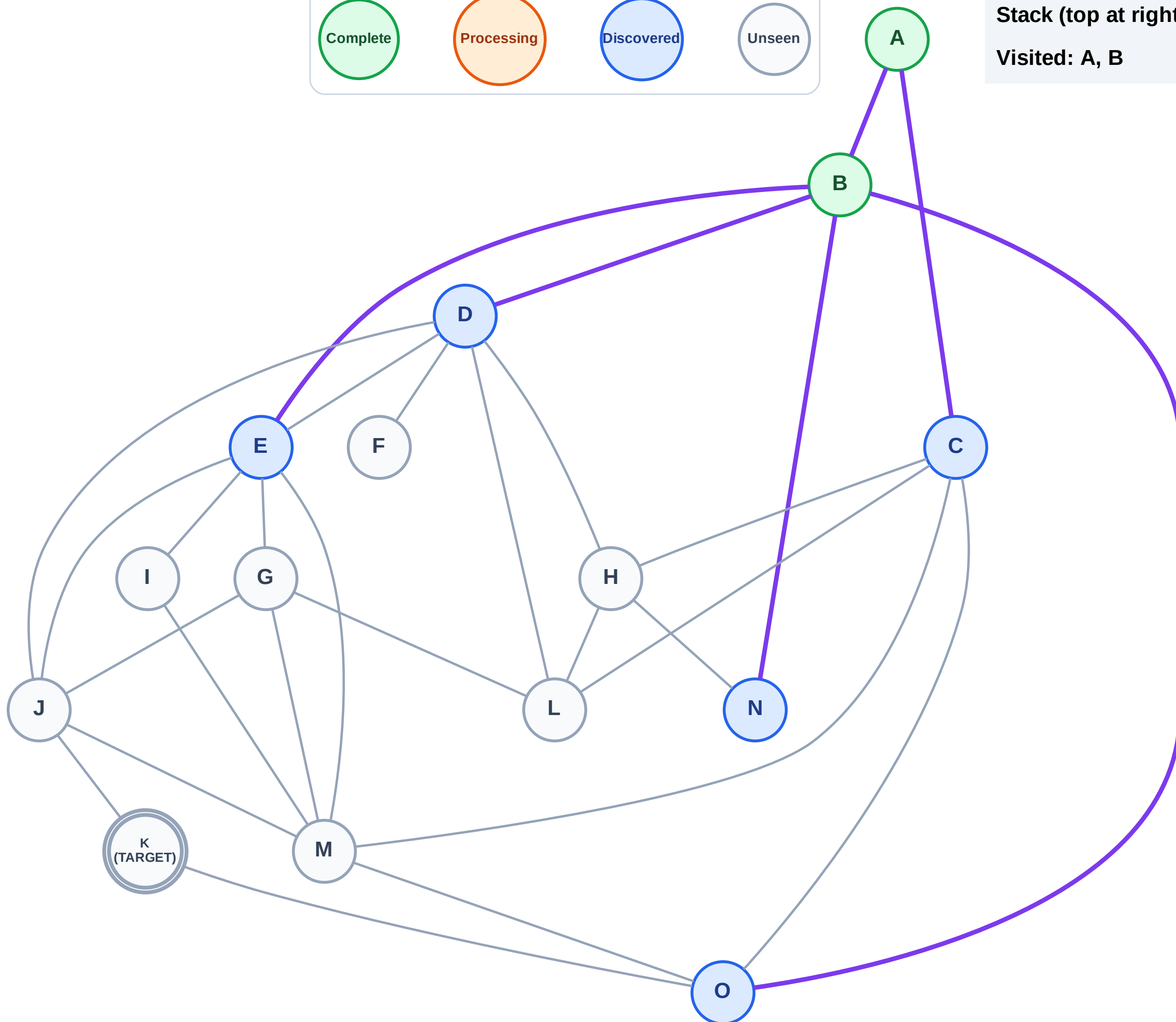
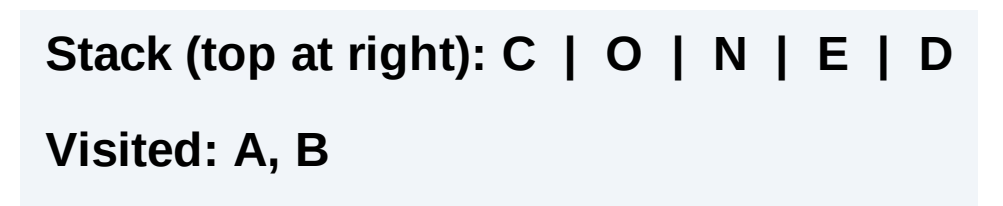


Stack (top at right): C

Visited: A



Step 5: Discover O, N, E, D from B



DFS Traversal

Step 6: Process node D

Legend

Complete

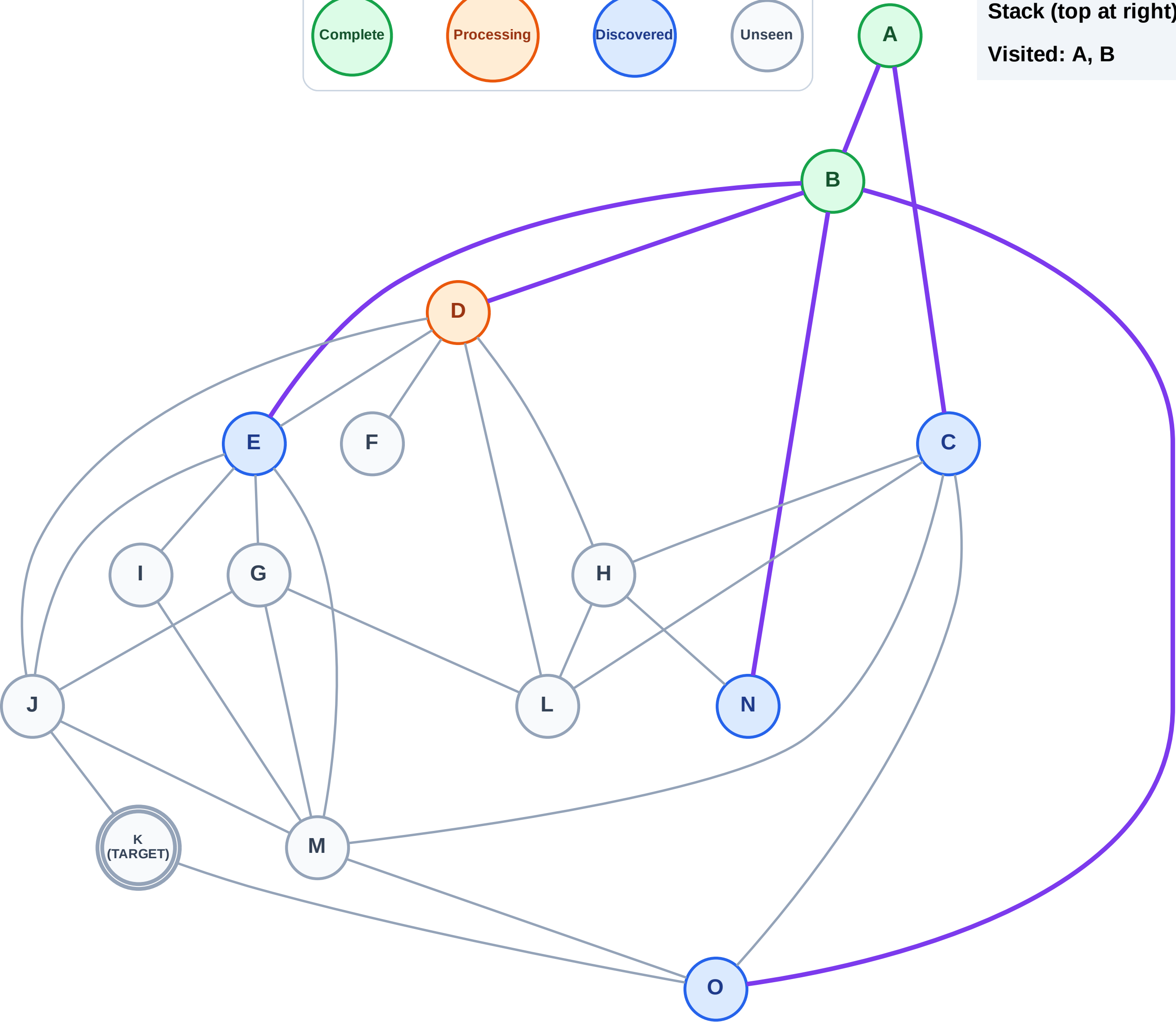
Processing

Discovered

Unseen

Stack (top at right): C | O | N | E

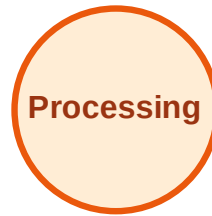
Visited: A, B



DFS Traversal

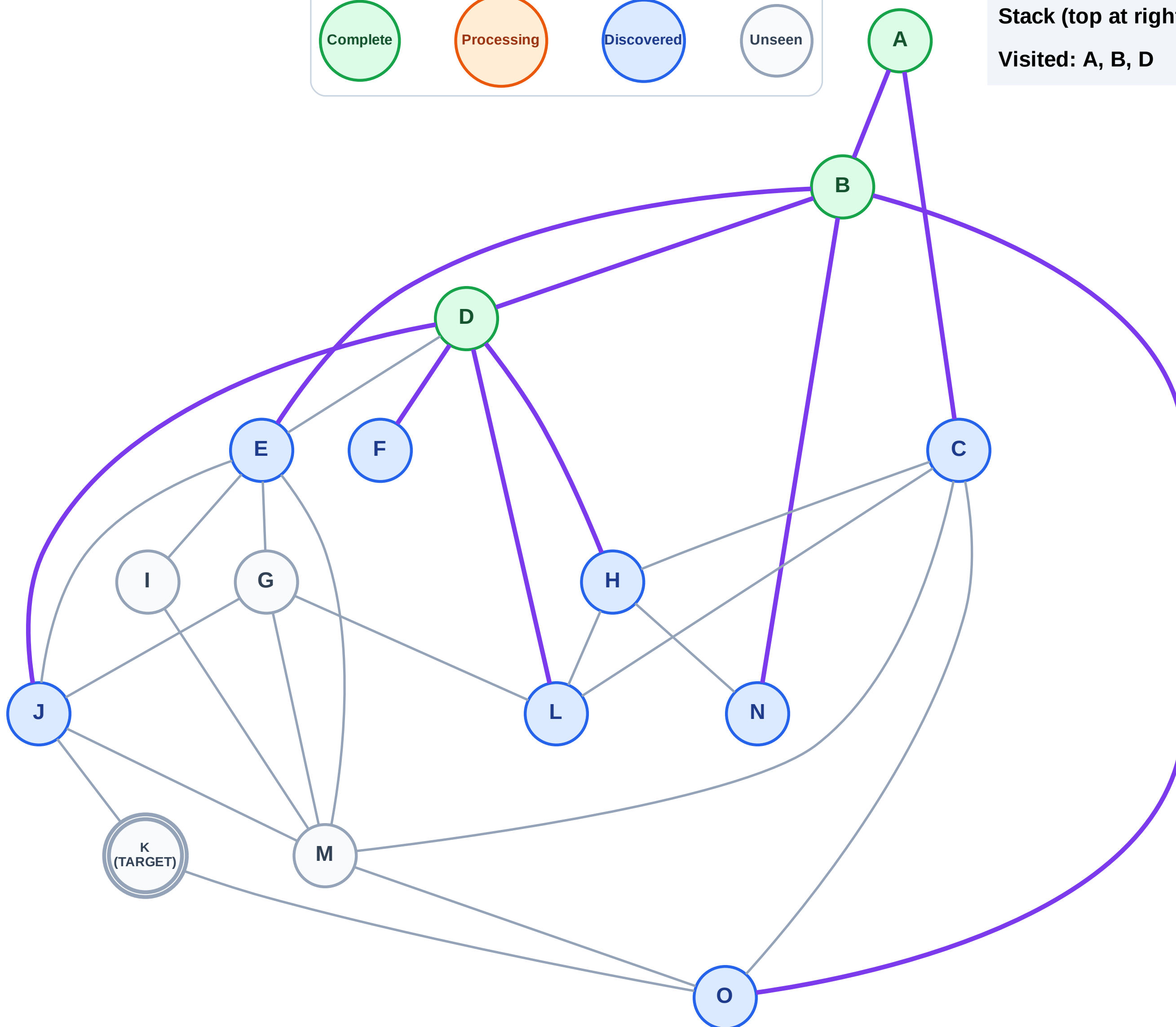
Step 7: Discover L, J, H, F from D

Legend



Stack (top at right): C | O | N | E | L | J | H | F

Visited: A, B, D



DFS Traversal

Step 8: Process node F

Legend

Complete

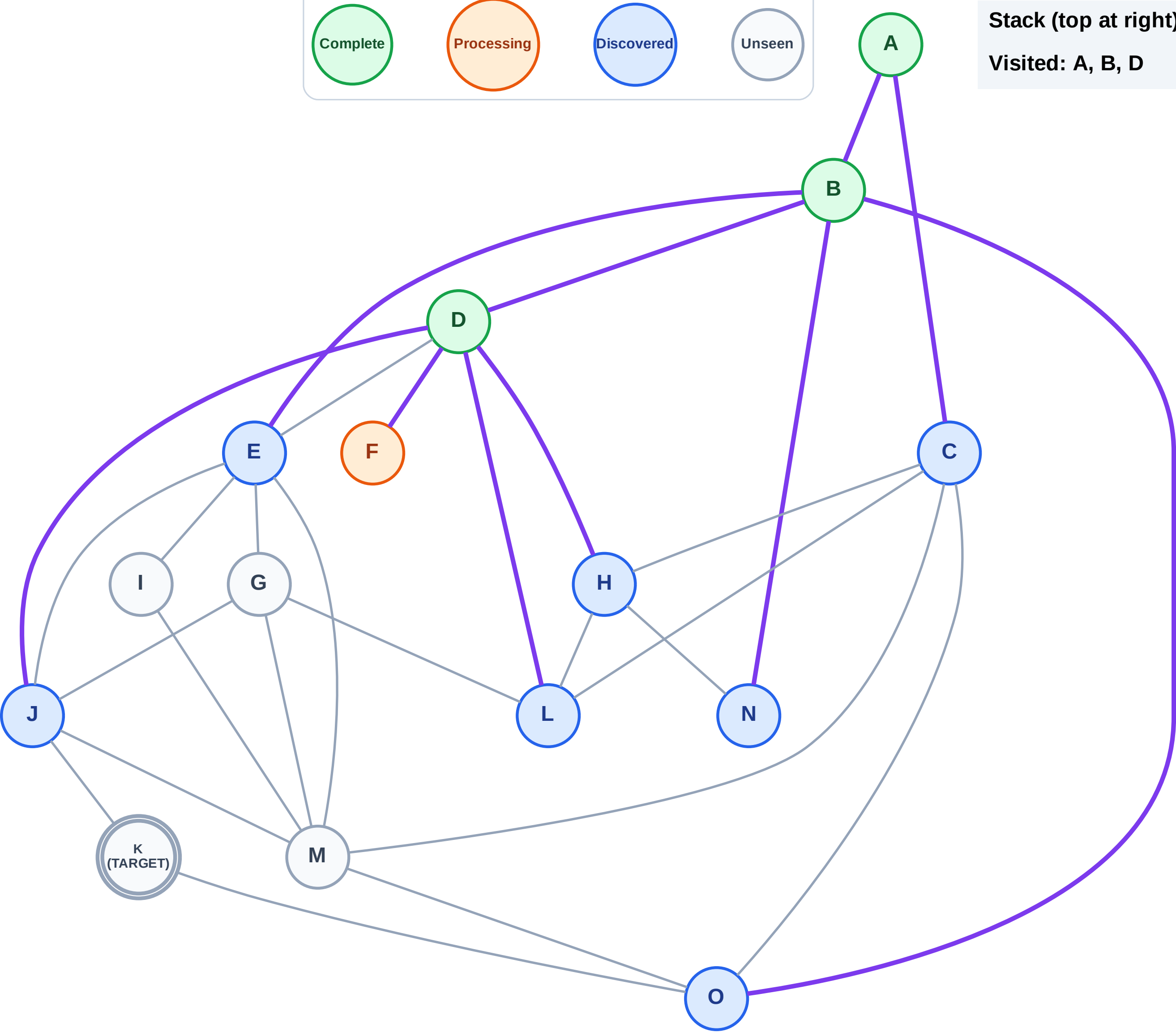
Processing

Discovered

Unseen

Stack (top at right): C | O | N | E | L | J | H

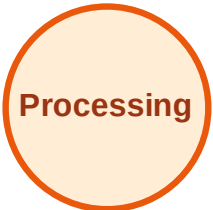
Visited: A, B, D



DFS Traversal

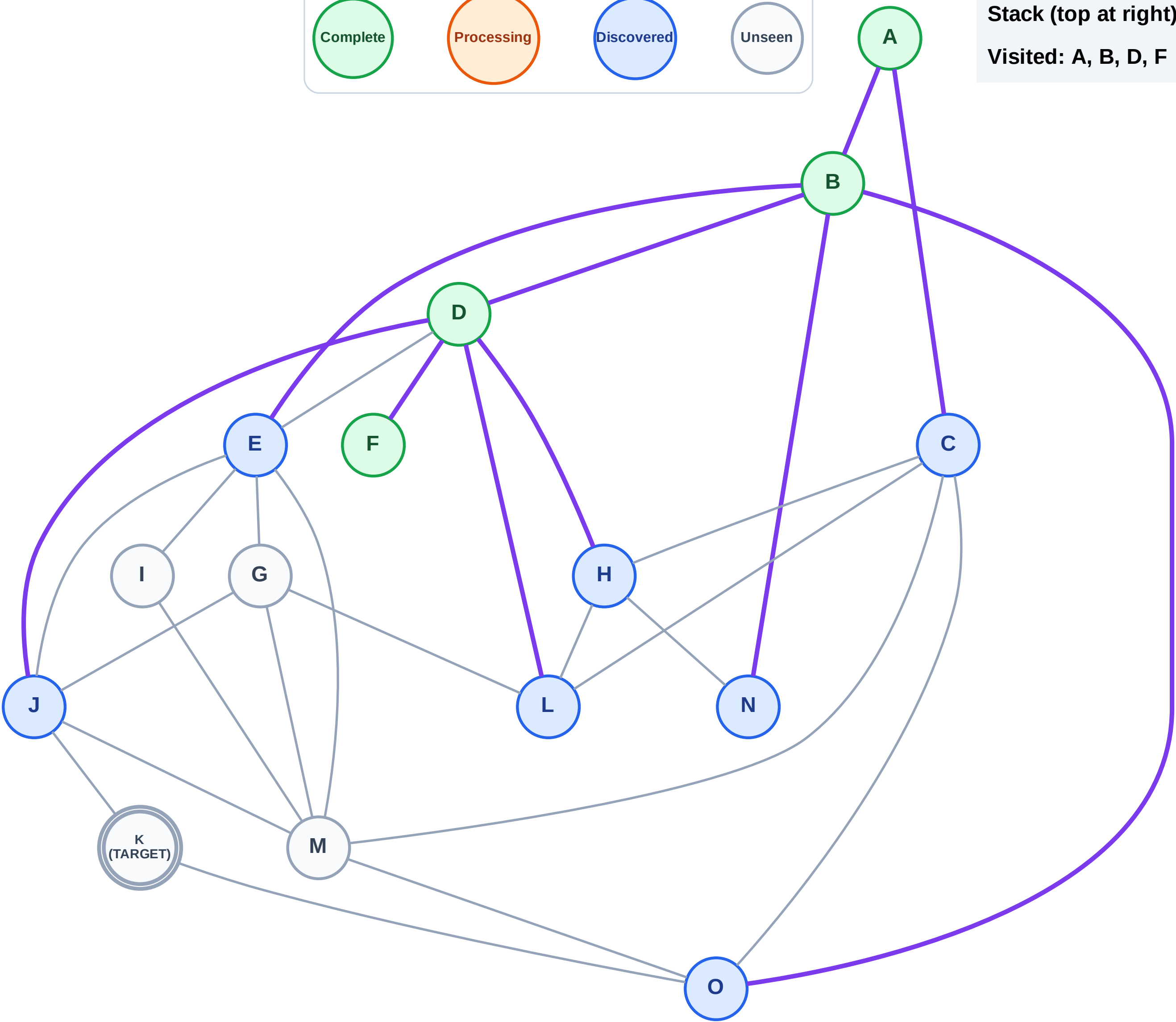
Step 9: No new neighbors from F

Legend



Stack (top at right): C | O | N | E | L | J | H

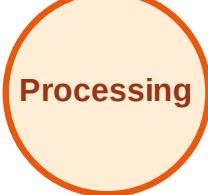
Visited: A, B, D, F



DFS Traversal

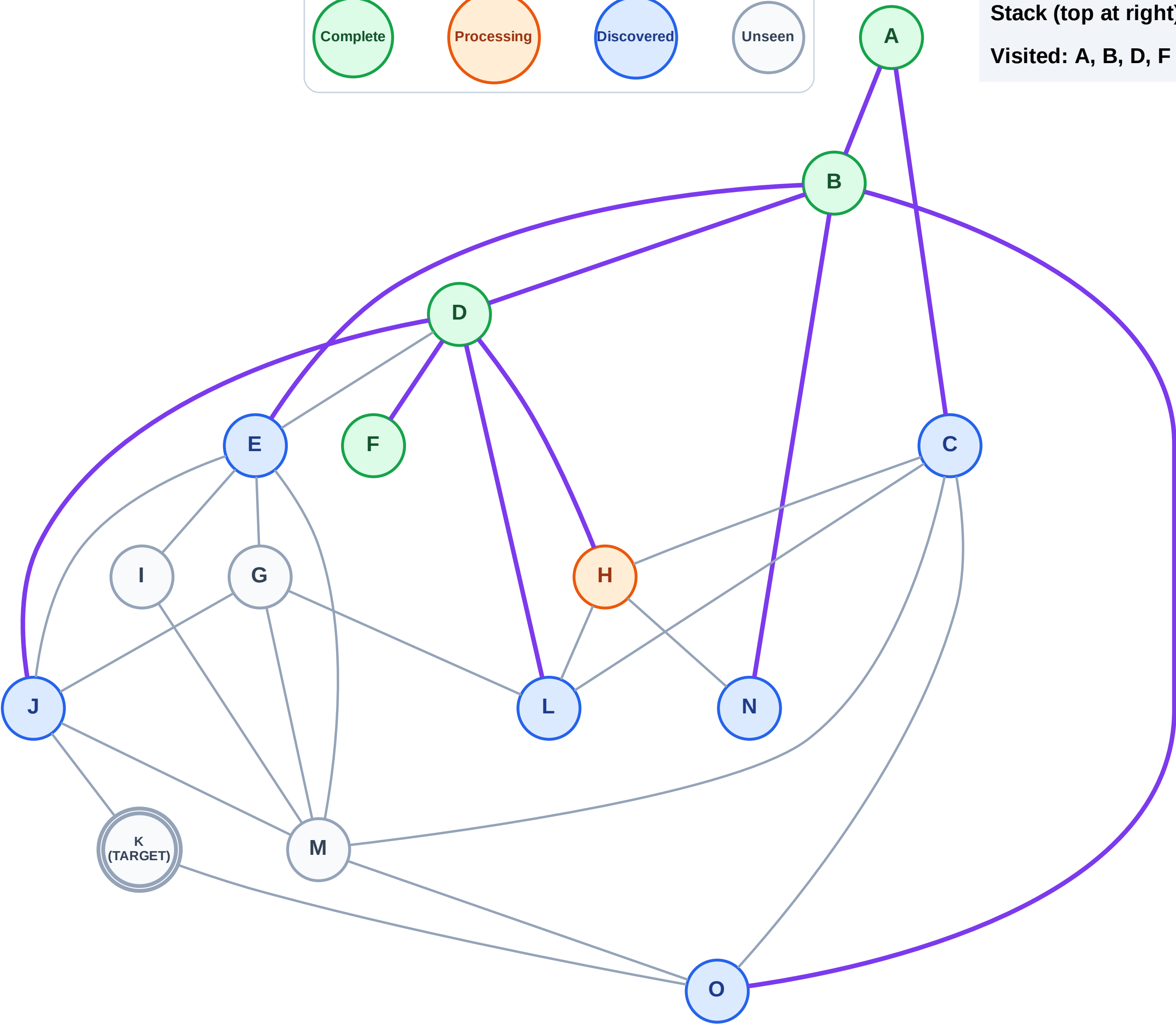
Step 10: Process node H

Legend



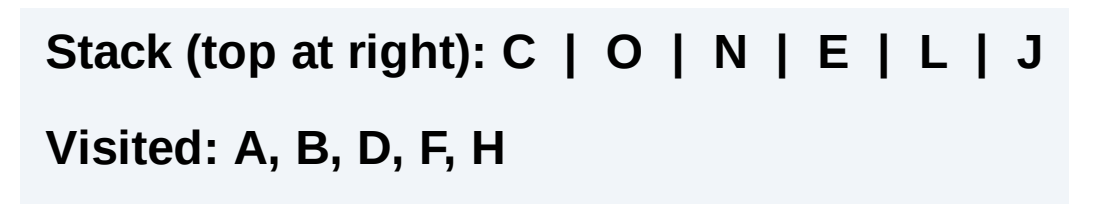
Stack (top at right): C | O | N | E | L | J

Visited: A, B, D, F



Step 11: No new neighbors from H

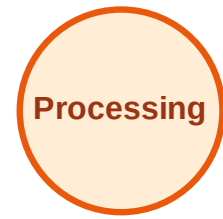
Step 11: No new neighbors from H



DFS Traversal

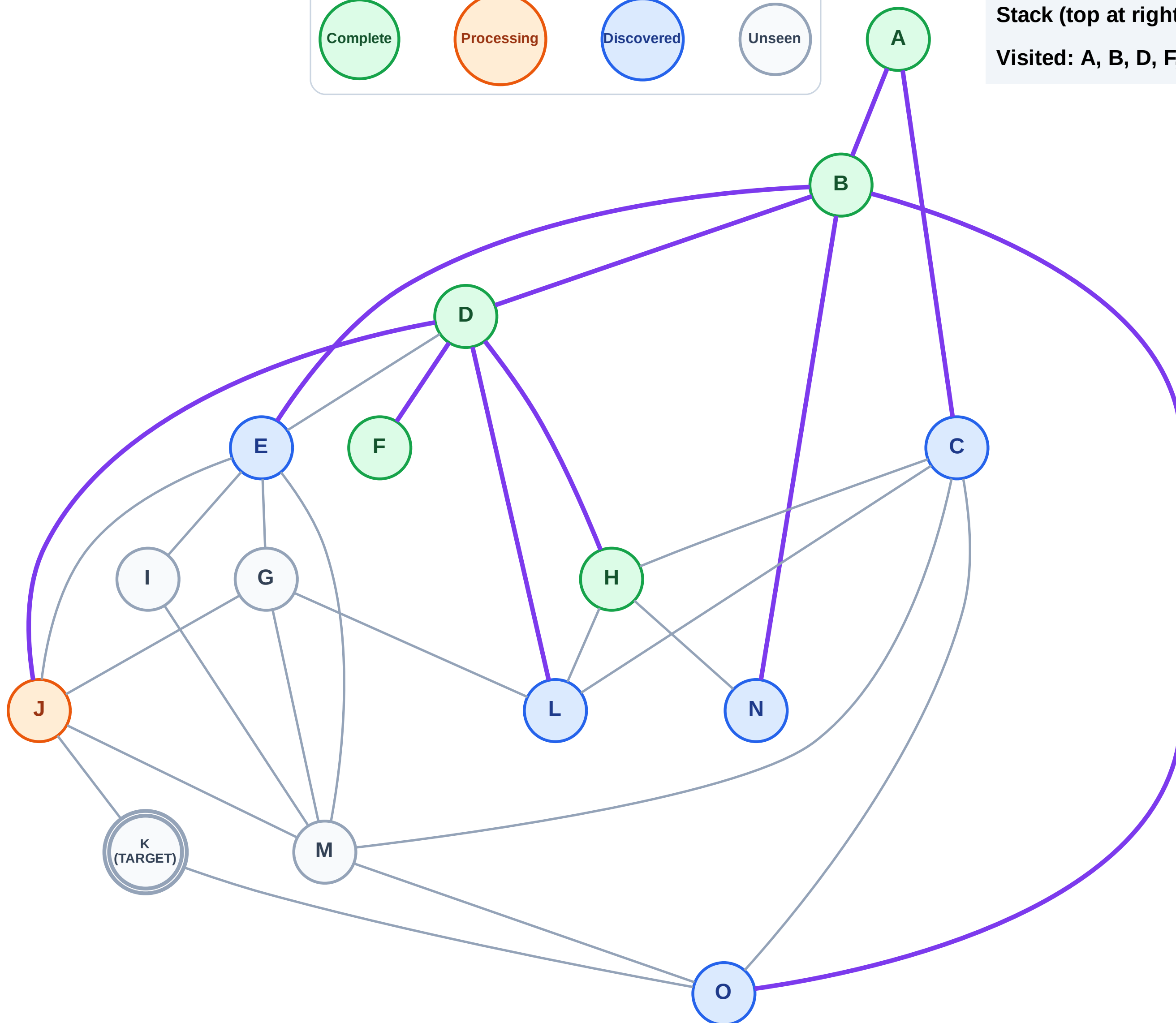
Step 12: Process node J

Legend



Stack (top at right): C | O | N | E | L

Visited: A, B, D, F, H



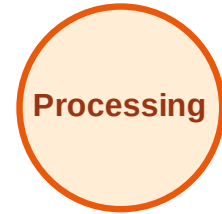
DFS Traversal

Step 13: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (4 hops): A → B → D → J → K

Legend



Complete



Processing



Discovered



Unseen

Stack (top at right): C | O | N | E | L | M

Visited: A, B, D, F, H, J, K

